

ELECTRIC MOTOR

How It Works



A Detailed Lesson for Grade 4 Students

Science · Energy Conversion · Electromagnetism

■ Subject: Science

■ Grade: 4

■ Duration: 45 min

■ Pages: 8

■ Learning Objectives

- Define an electric motor and explain its purpose.
- Identify the 3 main parts of an electric motor: Magnet, Coil of Wire, and Power Source.
- Describe the 4-step process by which a motor converts electrical energy into motion.
- Give real-life examples of devices that use electric motors.
- Explain the difference between like poles (repel) and unlike poles (attract) in magnets.

1. What Is an Electric Motor?

An **electric motor** is a device that **converts electrical energy into mechanical energy** (movement). Think of it like magic — electricity goes in, and spinning motion comes out! Every time you switch on a fan, start a washing machine, or play with a battery-operated toy car, an electric motor is silently doing its job.

■ Electrical Energy



■ Mechanical Energy (Spin)

■ Spot the Motor Activity

Device	Has a Motor?	Why?
Ceiling Fan	■ YES	Fan blades need to spin rapidly
Mixer-Grinder (Mixi)	■ YES	Blades grind food by spinning ~18,000 rpm
Book	■ NO	No moving parts; no energy conversion needed
Washing Machine	■ YES	Drum spins clothes to wash and dry them
Metro Train	■ YES	Traction motors move the entire train
Pencil	■ NO	Manual tool; requires no electrical energy

2. The Three Secret Ingredients

Every electric motor needs exactly **3 ingredients** to work its magic. Remember them using the shortcut **M-C-P: Magnet · Coil · Power!**

■ Part 1: The Magnet (The Pusher & Puller)

- A magnet can **push** and **pull** objects from a distance — without touching them.
- **Like poles REPEL** (push away): North vs North, South vs South.
- **Unlike poles ATTRACT** (pull together): North vs South.
- In a motor, the fixed magnet creates a magnetic field that interacts with the coil.
- *Real-life example:* Fridge magnets holding your drawings on the refrigerator door.

■ Part 2: The Coil of Wire (The Spinner / Armature)

- A **coil** is a long wire wound many times into loops — like winding a rubber band around a pencil.
- When electricity flows through the coil, it becomes an **electromagnet** — a temporary magnet.
- The more loops in the coil, the **stronger** the electromagnet produced.
- The coil sits between the poles of the permanent magnet and is free to spin.
- This spinning part is also called the **armature** or **rotor**.

■ Part 3: Electricity (The Power Source)

- Electricity is the **energy** that powers the entire motor.
- It can come from a **battery** (e.g., toy cars) or a **plug point / mains supply** (e.g., ceiling fans).
- When electricity flows into the coil, it creates the electromagnet.
- More electricity = stronger electromagnet = faster spinning.
- *Real-life example:* Turning a fan from Speed 1 to Speed 3 sends more current to the motor.

THE MOTOR FORMULA

■ Magnet + ■ Coil of Wire + ■ Power Source = ■■ Electric Motor

M — C — P = Motor!

3. How Does It Work? — The 4 Steps

Every electric motor follows the same **4-step process**, repeating continuously as long as the power is on. From a ceiling fan to a metro train — the same magic steps power them all!

1

Switch ON → Electricity Flows

When you flick the switch, electricity from the battery or plug point begins to flow through the coil of wire. Think of it like opening a tap — the current starts rushing in.

2

Coil Becomes an Electromagnet

As electricity flows through the coil, it creates a magnetic field around the wire. The coil is now a **temporary electromagnet** — it has its own North and South poles, just like a bar magnet. The more current flows, the stronger this electromagnet becomes.

3

Attraction & Repulsion → Coil Spins

The electromagnet's poles interact with the **fixed permanent magnet** around it. Unlike poles attract and like poles repel, creating a **push-pull force** (called torque). This force makes the coil rotate. A special part called the **commutator** reverses the current direction at just the right moment to keep the coil spinning in one direction.

4

Shaft Spins → Useful Work Done

The spinning coil is attached to a **shaft** (a metal rod). As the coil spins, so does the shaft. The shaft connects to the fan blade, wheel, or drum — whatever needs to move. The cycle repeats continuously as long as electricity flows!

■ Power ON



■ Electromagnet



■ Coil Spins



■ Work Done



■ Repeats

4. Real-Life Connections — Motor Safari! ■

Electric motors are **everywhere around us!** Let's go on a Motor Safari and find them in our daily lives. Once you know what to look for, you'll spot motors in every room of your house!

Device	Where Found	Speed / Fun Fact	Power Source
■ Ceiling Fan	Living Room / Bedroom	~400 rotations per minute!	Mains Plug
■ Mixer-Grinder (Mixer)	Kitchen	~18,000 rotations per minute!	Mains Plug
■ Washing Machine	Bathroom / Utility	Drum spins to clean clothes	Mains Plug
■ Delhi Metro / Train	Railway Stations	Traction motors — very powerful!	Electric Rail
■ Ola Electric Scooter	Roads / Streets	Replaces petrol engine with motor	Battery Pack
■ Battery-Operated Toy Car	Bedroom / Playroom	Small motor in each wheel	AA Batteries
— ■ Electric Pencil Sharpener	School / Study Room	Tiny motor spins the blade	Battery / USB
■ Water Motor Pump	Terrace / Basement	Pumps water to overhead tank	Mains Plug
■■ Exhaust Fan	Kitchen / Bathroom	Removes stale air from room	Mains Plug
■ Diwali Spinning Toys	Festival Celebrations	Festive motor magic!	Battery

■ **Did You Know?** The electric motors in the Delhi Metro are so powerful that they can accelerate a full train (with hundreds of passengers) from 0 to 80 km/h in under 30 seconds! The same principle — magnet + coil + electricity — powers everything from your toy car to an entire train.

5. Key Vocabulary

Term	Definition	Example
Electric Motor	A device that converts electrical energy into mechanical (spinning) movement.	<i>Ceiling fan, mixer-grinder</i>
Magnet	An object that creates a magnetic force — it can attract or repel other magnets.	<i>Fridge magnet, horseshoe magnet</i>
Electromagnet	A temporary magnet created when electricity flows through a coil of wire.	<i>Inside every motor</i>
Coil / Armature	A wire wound into loops that becomes an electromagnet when current flows through it.	<i>The spinning part of a motor</i>
Power Source	The source of electrical energy (battery or mains plug) that drives the motor.	<i>AA battery, wall socket</i>
Attract	When unlike magnetic poles pull towards each other (North and South).	<i>N pole pulls S pole</i>
Repel	When like magnetic poles push each other away (N-N or S-S).	<i>Two N poles push apart</i>
Commutator	A switch inside the motor that reverses current direction to keep the coil spinning.	<i>Inside DC motors</i>
Torque	The turning/spinning force produced by the interaction of magnets.	<i>What makes the coil rotate</i>
RPM	Rotations Per Minute — how many times something spins in one minute.	<i>Ceiling fan: ~400 RPM</i>

6. Activities & Assessment

■ Activity 1: Let's Be the Motor!

Instructions: Stand up! Stretch your arms out wide — you are now a coil of wire. When your teacher says 'Switch ON', start spinning slowly (low electricity). When your teacher says 'More Power!', spin faster. When your teacher says 'Switch OFF', stop. Feel the science in your body!

Discussion: What happened to your speed when more power was added? What does this tell us about the relationship between electricity and motor speed?

■ Activity 2: Motor Detective Mission (Homework)

Mission: Go home and be a Motor Detective! 1. Walk through every room of your house. 2. Find ALL devices that use an electric motor — make a list. 3. Draw ONE motor-powered device and **label its three parts:** Magnet, Coil/Armature, Power Source. 4. Bring your drawing to class tomorrow and share your findings!

■ Check Your Understanding

Q1: Name TWO things in your home that use an electric motor.

Answer: Any two from: ceiling fan, mixer-grinder, washing machine, exhaust fan, water pump, toy car, etc.

Q2: What are the 3 main parts of an electric motor?

Answer: Magnet, Coil of Wire (Armature), and Power Source (M-C-P)

Q3: What type of energy goes INTO the motor?

Answer: Electrical Energy

Q4: What type of energy comes OUT of the motor?

Answer: Mechanical Energy (spinning/movement)

Q5: True or False: A book uses an electric motor. Explain your answer.

Answer: FALSE — A book has no moving parts and requires no electrical energy. It is a static object.

Q6: What happens if you add MORE electricity to a motor?

Answer: The electromagnet becomes stronger, the forces increase, and the motor spins FASTER.

Q7: What is an electromagnet? How is it different from a permanent magnet?

Answer: An electromagnet is a temporary magnet made by passing electricity through a coil. Unlike a permanent magnet, it can be switched on and off.

7. Quick Recap & Big Ideas

BIG IDEA

■ A motor turns electricity into spinning movement!

THREE PARTS

■ Magnet + Coil + Power Source = Electric Motor (M-C-P)

4

FOUR STEPS

■ Power ON → Electromagnet → Spin → Useful Work → Repeat!

ENERGY

■ Electrical Energy (input) → Mechanical Energy (output)

EVERYWHERE

■ Fans, mixers, trains, scooters, toys — motors power our world!

■ **Coming Up Next:** How does a **Generator** work? (Hint: It's the opposite of a motor — it turns *movement* into *electricity*!)